

Project Data Dictionary

This document defines the columns used in the project's analytical views. The definitions are kept consistent throughout the analysis to ensure that business questions and SQL queries use the same interpretation of each metric.

The analytical layer consists of five primary views:

View	Analytical Level	Purpose
vw_order_summary	Order	Order-level sales, payment, review and delivery analysis
vw_product_sales	Order Item	Product, seller, category and item-level sales analysis
vw_customer_summary	Customer	Customer purchasing behaviour and value
vw_seller_summary	Seller	Seller sales, volume, satisfaction and delivery performance
vw_category_summary	Category	Category sales, volume, satisfaction and delivery performance

1. vw_order_summary

Grain: One row per order.

This is the primary view for order-level, revenue, customer experience and delivery analysis.

Order & Customer

Column	Definition
order_id	Unique identifier of the order
customer_id	Identifier of the customer record associated with the order
customer_unique_id	Persistent customer identifier used to identify the same underlying customer across customer records
customer_city	City associated with the customer
customer_state	Brazilian state associated with the customer
customer_zip_code_prefix	Customer's ZIP-code prefix

Important distinction: `customer_id` identifies a customer record, whereas `customer_unique_id` identifies the underlying customer. Therefore, customer-level behavioural analysis should use `customer_unique_id`.

Order Status

Column	Definition
<code>order_status</code>	Current/final status recorded for the order

Observed statuses include: delivered, shipped, canceled, unavailable, invoiced, processing, created, approved.
For delivery-performance analysis, orders are generally restricted to: `order_status = 'delivered'`

Order Timestamps

Column	Definition
<code>order_purchase_timestamp</code>	Date and time when the customer placed the order
<code>order_approved_at</code>	Date and time when the order was approved
<code>order_delivered_carrier_date</code>	Date and time when the order was handed over to the carrier
<code>order_delivered_customer_date</code>	Date and time when the order was delivered to the customer
<code>order_estimated_delivery_date</code>	Estimated/promised delivery date

Order-Level Quantities

Column	Definition
<code>total_items</code>	Total number of order items contained in the order

Financial Metrics

Column	Definition
<code>product_value</code>	Sum of product prices across all items in the order
<code>freight_value</code>	Sum of freight charges across all items in the order
<code>total_payment_value</code>	Total recorded payment value associated with the order
<code>max_installments</code>	Maximum number of payment installments associated with the order
<code>order_value</code>	Order value calculated as <code>product_value</code> + <code>freight_value</code>

- `product_value` = $\sum$ item price
- `freight_value` = $\sum$ item freight
- `order_value` = `product_value` + `freight_value`

`total_payment_value` is kept as a separate metric because it originates from the payment records rather than being calculated from order items.

Review Metrics

Column	Definition
average_review_score	Average review score associated with the order
review_count	Number of review records associated with the order

Review scores range from 1 to 5, where higher values indicate better customer satisfaction.

Derived Logistics Metrics

Column	Definition
approval_time_hours	Time between order purchase and order approval, measured in hours
carrier_processing_days	Time between order purchase and carrier handoff, measured in days
delivery_time_days	Time between order purchase and actual customer delivery, measured in days
estimated_delivery_days	Time between order purchase and estimated delivery date, measured in days
delivery_delay_days	Difference between actual delivery date and estimated delivery date, measured in days

Key Metric (delivery_delay_days): actual delivery date – estimated delivery date
< 0 → delivered early | = 0 → delivered on time | > 0 → delivered late

2. vw_product_sales

Grain: One row per order item.

This view connects individual sales transactions with their corresponding products, categories and sellers.

Identifiers & Product Information

Column	Definition
order_id	Order containing the item
order_item_id	Item sequence number within the order
product_id	Unique product identifier
seller_id	Unique seller identifier
product_category	Standardized English product category
product_name_length	Number of characters in the product name
product_description_length	Number of characters in the product description
product_photos_qty	Number of product photos
product_weight_g	Product weight in grams

product_length_cm	Product length in centimetres
product_height_cm	Product height in centimetres
product_width_cm	Product width in centimetres

Seller Location & Item-Level Financial Metrics

Column	Definition
seller_zip_code_prefix	Seller's ZIP-code prefix
seller_city	Seller's city
seller_state	Seller's Brazilian state
price	Selling price of the individual item
freight_value	Freight charge associated with the individual item
item_total_value	Total item value calculated as price + freight_value

3. vw_customer_summary

Grain: One row per customer_unique_id.

This view is used to analyse customer behaviour, purchasing frequency, monetary value and experience.

Customer Identity, Location & Order Activity

Column	Definition
customer_unique_id	Persistent unique identifier for the customer
customer_city	Customer's city
customer_state	Customer's state
total_orders	Number of distinct orders associated with the customer
delivered_orders	Number of distinct delivered orders
canceled_orders	Number of distinct canceled orders

Customer frequency classification:

total_orders = 1 → One-time customer

total_orders > 1 → Repeat customer

Customer Monetary, Lifecycle & Experience

Column	Definition
total_spent	Total order_value attributed to the customer's orders
average_order_value	Average order_value across the customer's orders

<code>first_purchase</code>	Date and time of the customer's earliest purchase
<code>last_purchase</code>	Date and time of the customer's most recent purchase
<code>average_review_score</code>	Average review score across the customer's reviewed orders
<code>average_delivery_days</code>	Average delivery time for the customer's delivered orders
<code>average_delivery_delay_days</code>	Average delivery delay for the customer's delivered orders

| 4. vw_seller_summary

Grain: One row per seller.

This view evaluates sellers across three dimensions: Commercial performance + customer satisfaction + logistics performance.

Seller Attributes & Metrics

Column	Definition
<code>seller_id</code>	Unique seller identifier
<code>seller_city</code>	Seller's city
<code>seller_state</code>	Seller's Brazilian state
<code>total_orders</code>	Number of distinct orders containing products sold by the seller
<code>total_items_sold</code>	Total number of individual items sold
<code>unique_products_sold</code>	Number of distinct products sold by the seller
<code>total_product_sales</code>	Sum of product prices generated by the seller
<code>total_freight</code>	Total freight value associated with the seller's items
<code>total_sales_value</code>	<code>total_product_sales</code> + <code>total_freight</code>
<code>average_item_price</code>	Average selling price per item
<code>average_order_value</code>	Average order-level value associated with the seller's orders
<code>average_review_score</code>	Average customer review score associated with the seller's orders
<code>average_delivery_days</code>	Average delivery time for the seller's delivered orders
<code>average_delivery_delay_days</code>	Average delivery delay for the seller's delivered orders

| 5. vw_category_summary

Grain: One row per product category.

This view evaluates categories according to sales, volume, customer satisfaction and delivery performance.

Category Attributes & Metrics

Column	Definition
product_category	Standardized product category
total_orders	Number of distinct orders containing products from the category
total_items_sold	Number of individual items sold in the category
unique_products	Number of distinct products belonging to the category
unique_sellers	Number of distinct sellers selling products in the category
total_product_sales	Sum of product prices generated by the category
total_freight	Total freight value associated with the category
total_sales_value	total_product_sales + total_freight
average_item_price	Average product selling price
average_review_score	Average review score associated with the category
average_delivery_days	Average delivery time for orders containing products from the category
average_delivery_delay_days	Average delivery delay for orders containing products from the category

| Core Metrics Used Throughout the Analysis

The majority of the project's business questions rely on a smaller set of core metrics.

Revenue & Sales

- `order_value`
- `product_value`
- `freight_value`
- `total_payment_value`
- `price`
- `item_total_value`
- `total_product_sales`
- `total_sales_value`
- `total_spent`

Customer Behaviour

- `customer_unique_id`
- `total_orders`
- `delivered_orders`
- `canceled_orders`
- `total_spent`
- `average_order_value`
- `first_purchase`
- `last_purchase`

Customer Experience

- `average_review_score`
- `review_count`

Seller Performance

- `seller_id`
- `total_orders`
- `total_items_sold`
- `unique_products_sold`
- `total_product_sales`
- `total_sales_value`

Category Performance

- `product_category`
- `total_orders`
- `total_items_sold`
- `unique_products`
- `unique_sellers`
- `total_sales_value`

Delivery Performance & Time

- `delivery_time_days`, `estimated_delivery_days`,
`delivery_delay_days`
- `average_delivery_days`, `average_delivery_delay_days`
- `order_purchase_timestamp`, `order_approved_at`
- `order_delivered_carrier_date`, `order_delivered_customer_date`
- `order_estimated_delivery_date`

Analytical Grain Convention

A critical principle throughout this project is that the same metric can have a different interpretation depending on the analytical grain of the view. This distinction is essential when constructing SQL queries because aggregating a metric at the wrong level can lead to double counting or misleading results.

View	Grain	Example
<code>vw_order_summary</code>	Order	<code>order_value</code> = value of an individual order
<code>vw_product_sales</code>	Order Item	<code>item_total_value</code> = value of an individual item including freight
<code>vw_customer_summary</code>	Customer	<code>total_orders</code> = number of orders placed by a customer
<code>vw_seller_summary</code>	Seller	<code>total_orders</code> = distinct orders containing products sold by the seller
<code>vw_category_summary</code>	Category	<code>total_orders</code> = distinct orders containing products from the category